EV charging in Rebana: who can charge, who cannot, and is the grid ready?

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All layers are also available in ur-scape, the workshop's shared visualisation platform

Rebana charges the cars passing through, not the cars of people who live there

73 %

of Rebana's charging energy is sold at 19 toll-road rest areas

29 %

of residents live within a 10-minute drive of a charger (Jabodetabek 76 %)

1.04 M

residents (11 %) are more than 30 minutes from any charger

−15 MVA

headroom at Sunyaragi substation, which supplies Kota Cirebon

Access is uneven and follows density, not need. The sparsest fifth of villages has 5 % of residents within 10 minutes; the densest fifth has 70 %.

Grid capacity exists, but in the wrong places. Haurgeulis, Cikedung and Arjawinangun Baru substations have 56–69 % headroom and almost no chargers.

Because Rebana is still being built, siting can be decided before demand forms. The village travel-time method becomes an ex-ante planning tool, not an audit.

Seven regencies, 9.2 million people, and a metropolitan area still under construction

Area	Population (M)	Industrial zones (KPI)	Investment 2025 (Rp T)	Chargers per 100k
Kab. Cirebon	2.45	1	4.01	2.49
Kab. Indramayu	1.95	6	3.35	1.06
Kab. Subang	1.66	3	18.2	1.20
Kab. Majalengka	1.36	3	3.36	1.74
Kab. Kuningan	1.24	0	–	1.16
Kab. Sumedang	1.17	0	5.63	1.30
Kota Cirebon	0.36	0	–	1.96

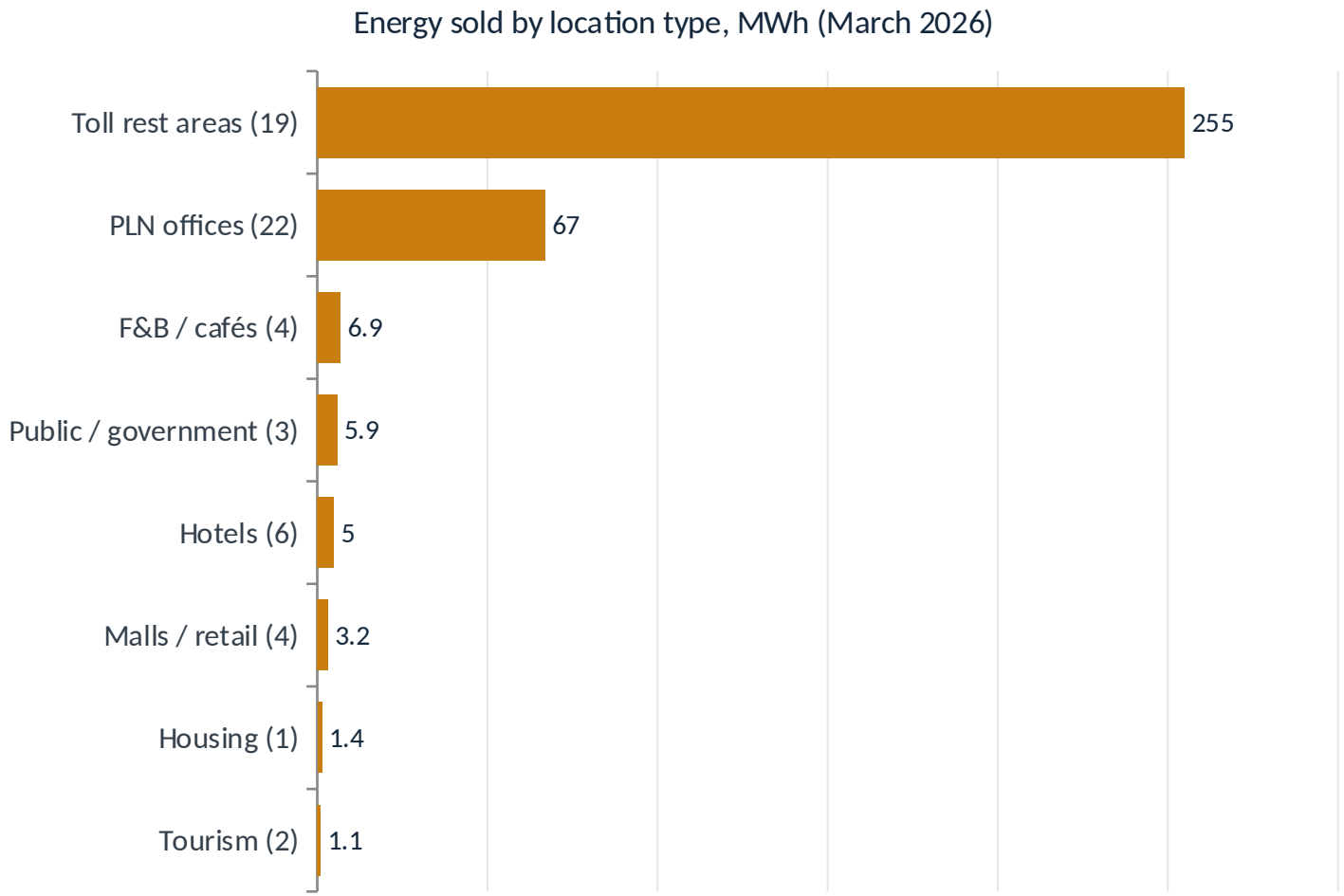
Rp 36.7 T
investment in 2025 (BKPM via BP Rebana)

13 industrial zones
largest: Cipali Subang Barat 10,408 ha, Losarang 6,170 ha

1.78 M jobs targeted by 2030
plus Kertajati airport and Patimban port

Sources: Kontur 2023, BPS 2024, BKPM/BP Rebana; chargers: PLN Charge.IN

Charging today follows the toll corridor, not the towns



13,691

charging sessions in March 2026, 13.6 % of West Java

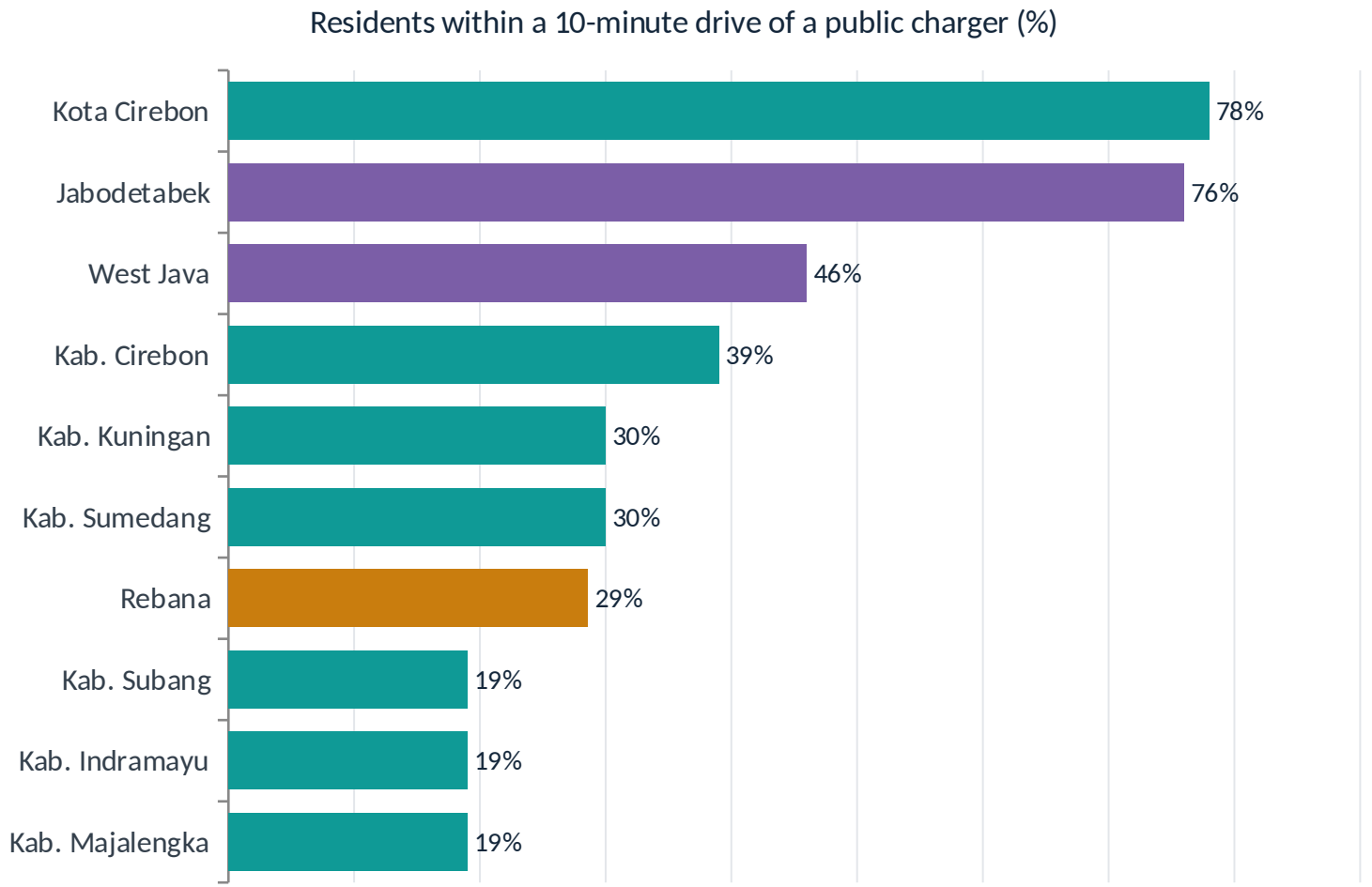
60 %

of sessions at chargers of 100 kW or more

< 100

EVs registered by Rebana residents

Fewer than three in ten residents can reach a charger in 10 minutes



How it is measured

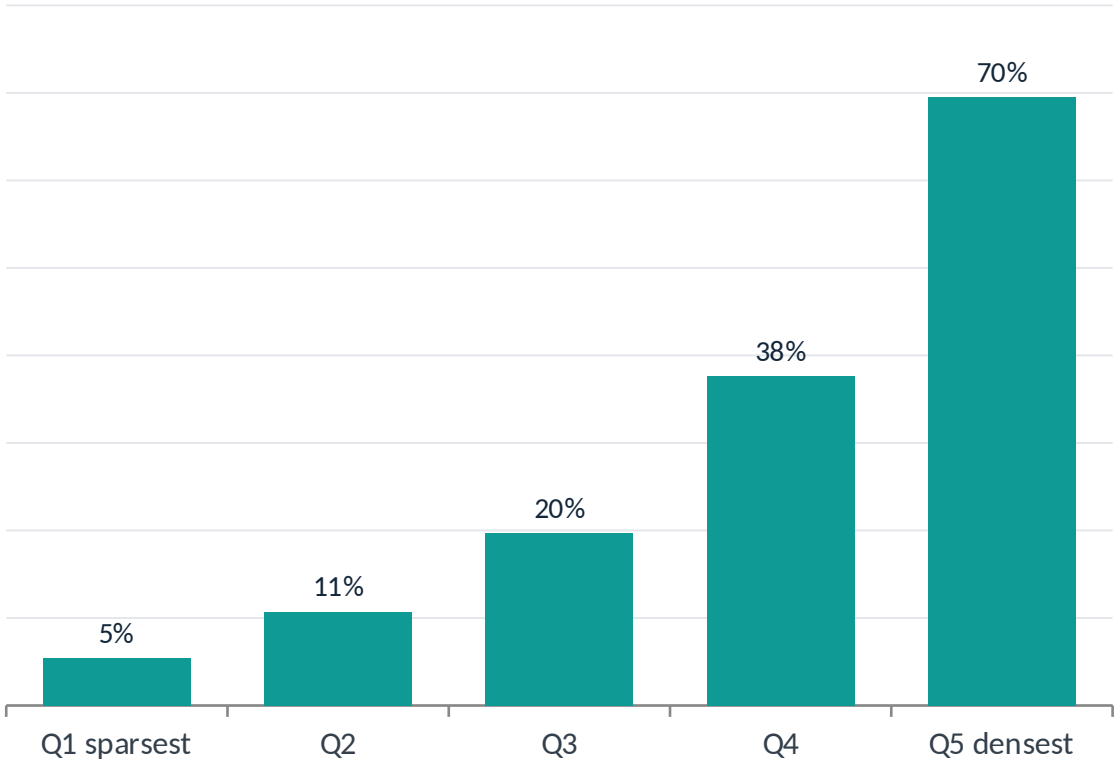
Car travel time on the OSM road network (r5r village-to-village matrix, 60-minute cap, by Alyas Widita) from each of 1,981 Rebana villages to the nearest village with a public charger (1,556 sites, PLN Charge.IN June 2026), weighted by Kontur 2023 population.

Within 30 minutes: 88.8 %

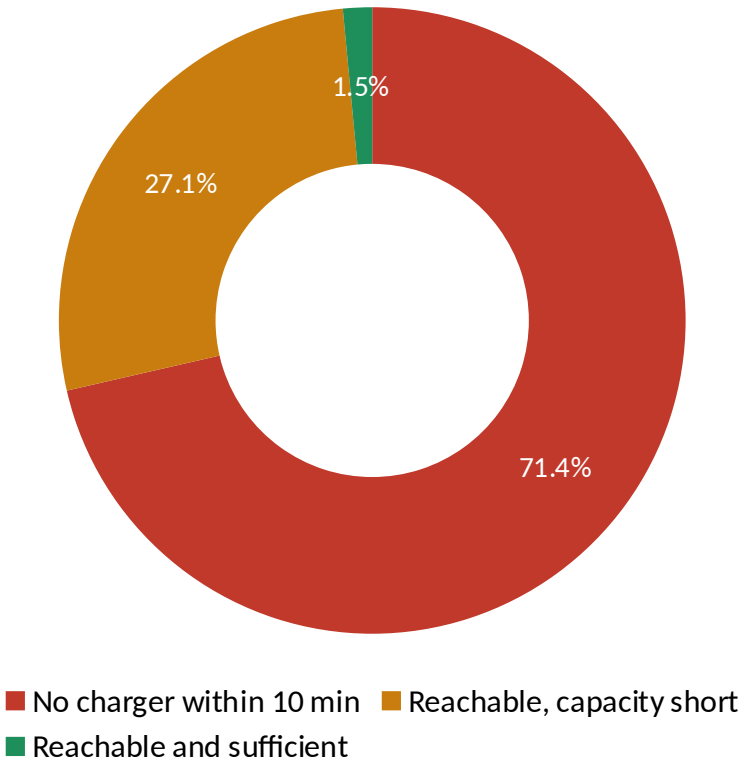
Kuningan and Sumedang each have about 250,000 residents beyond 30 minutes.

The gap is rural versus urban, and the service ladder stops at the first rung

Within 10 minutes, by village density fifth

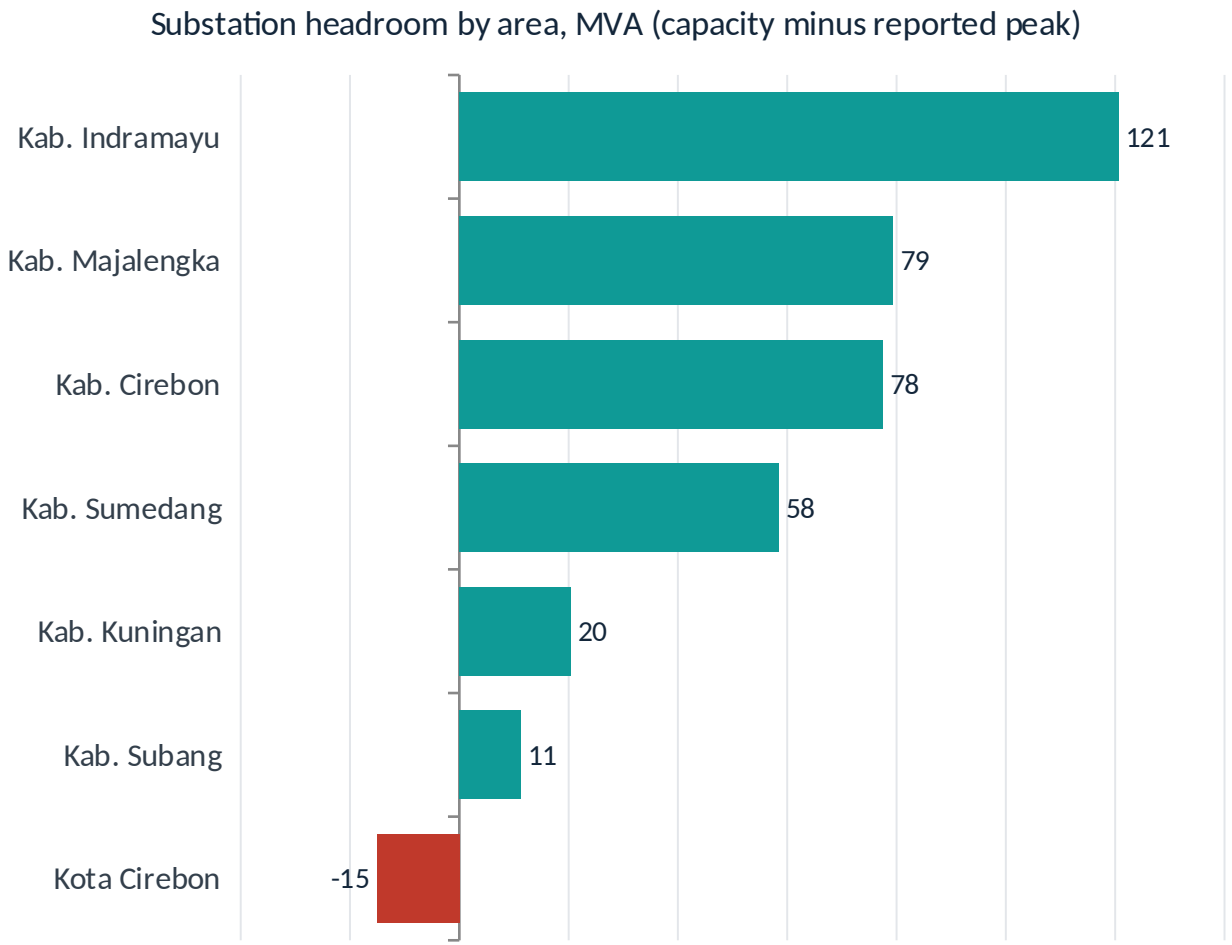


Service ladder: share of Rebana residents



Concentration index of 15-minute access against density is 0.31; against spending per person only 0.08. Inequality here is rural–urban, not rich–poor, and can still be prevented. Sufficiency = E2SFCA charger kW within 10 minutes ≥ 0.03 kW per household.

Grid headroom exists where chargers do not



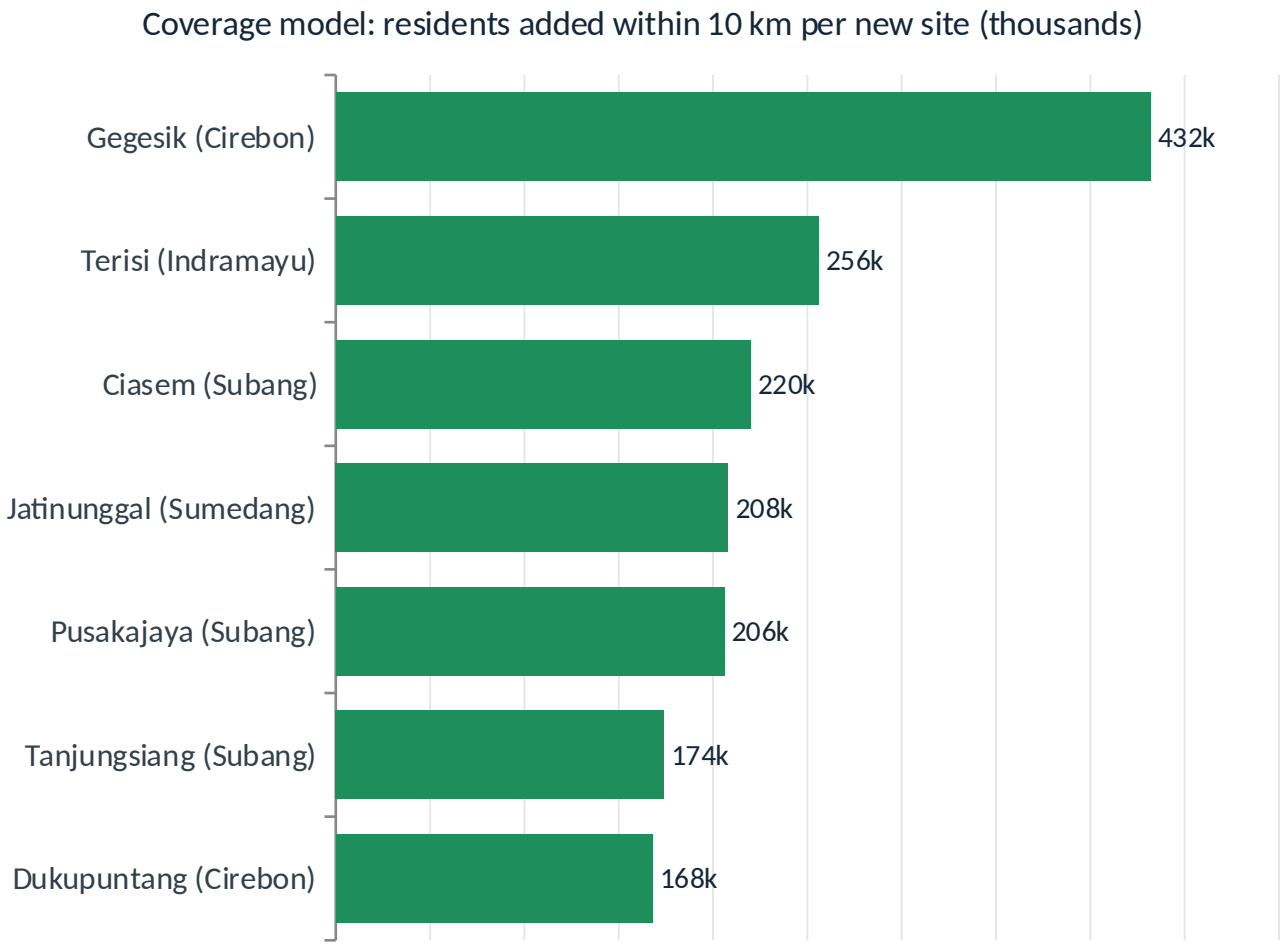
Risk 1 · saturated towns

Sunyaragi (Kota Cirebon) is at 86 % of peak, Subang at 89.5 %, Indramayu and Jatibarang above 80 %. Fast chargers there add midday peak to full substations.

Risk 2 · idle capacity

Haurgeulis (69 %), Cikedung (56 %) and Arjawinangun Baru (56 %) serve sub-districts with no charger. Pair new rural sites with these, and daytime solar tariffs in towns.

Two siting models point to the same three places



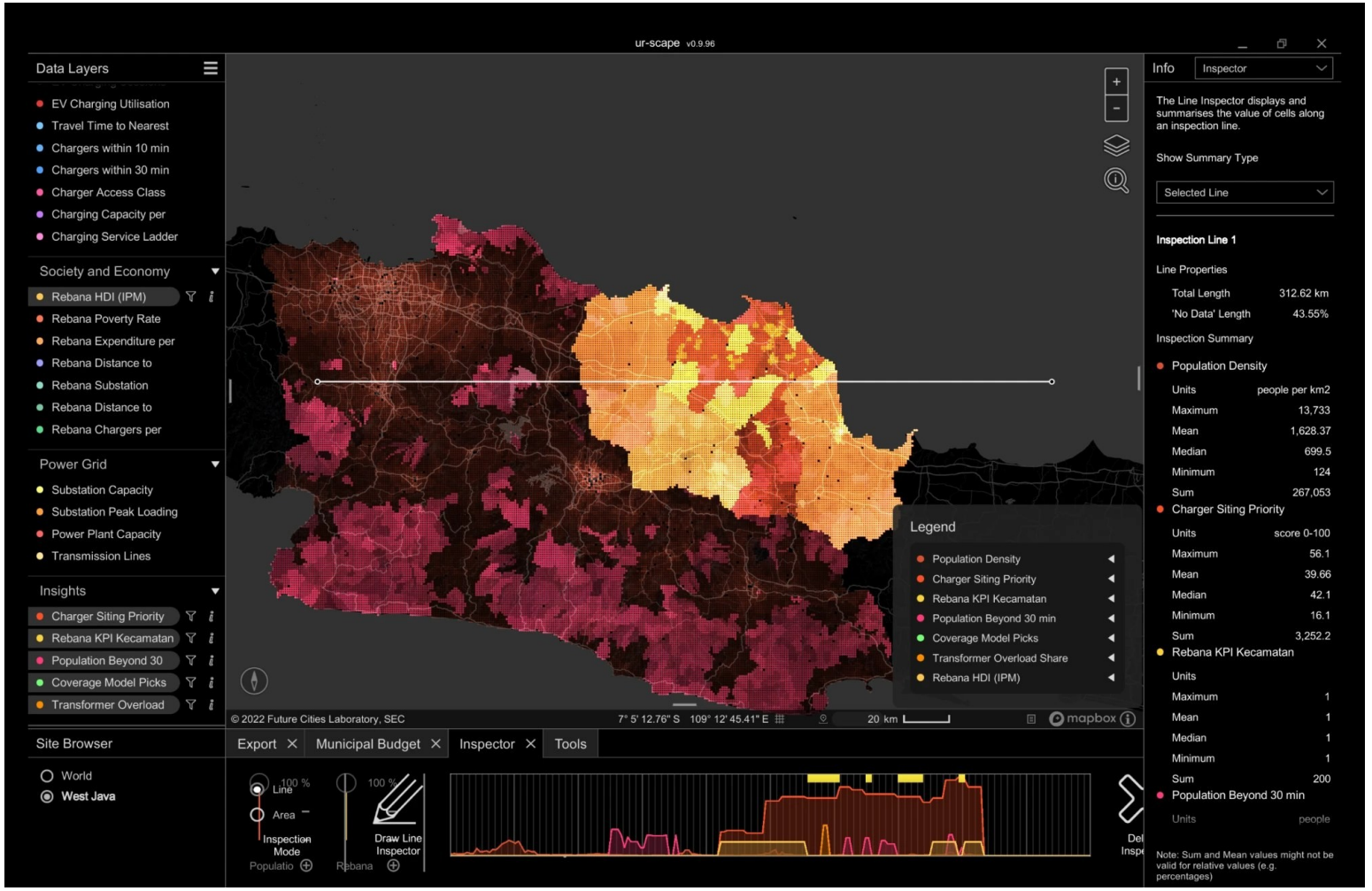
Model A · siting priority score

0.5 × need (travel time, distance) + 0.25 × demand (log population) + 0.25 × readiness (substation headroom). Top: Pusakanagara 67.0, Cibingbin 65.9, Krangkeng 65.9, Losari 65.2, Gegesik 63.2.

Both models meet at Gegesik, Terisi and Pusakanagara–Pusakajaya

One 4-gun station in Gegesik adds 432,000 residents to the 10 km radius, more than all new chargers in 2026 combined. Twelve sites lift coverage from 67 % to about 90 %.

The same evidence in ur-scape, the workshop's shared platform



28 layers, one site

Site "West Java", 0.5 arc-minute cells (about 0.9 km), no QGIS needed.

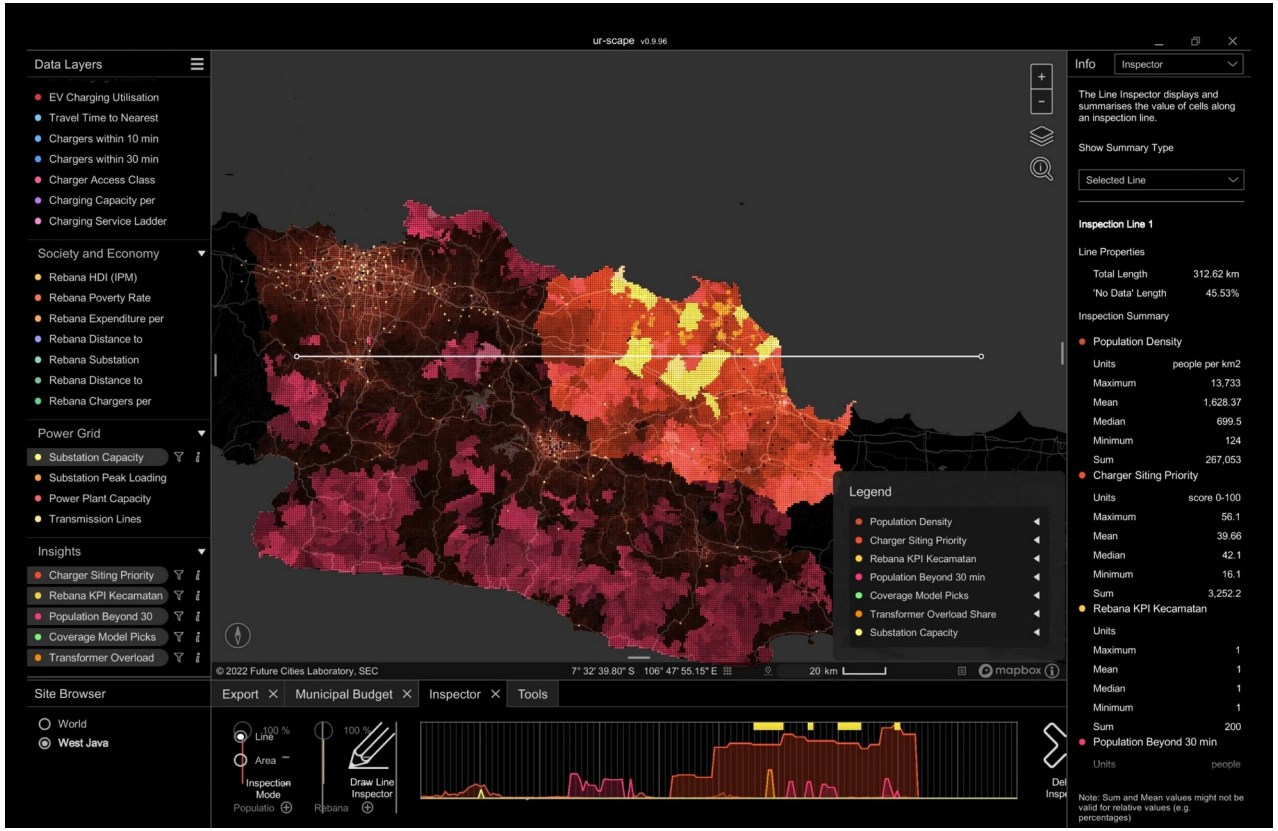
EV Charging
stations, energy, utilisation, travel time, access class, service ladder

Society and Economy
density, Rebana HDI, poverty, expenditure, distances

Power Grid
substations, peak loading, power plants, transmission

Insights
siting priority, access gap, coverage picks, KPI kecamatan, transformer overload

Reading one line across Rebana with the ur-scape Inspector



Layer along the 313 km line	Min	Median	Mean	Max
Population density (people/km²)	124	700	1,628	13,733
Charger siting priority (0–100)	16.1	42.1	39.7	56.1

What the line shows

- Density along the line is skewed: median 700 but mean 1,628 people per km². A few town cells hold most residents.
- Siting priority ranges from 16 to 56 (median 42). The profile graph under the map shows where along the line it rises.
- The line crosses industrial-zone (KPI) sub-districts at 200 sample points.
- 42–53 % of the line has no data because Rebana-only layers stop at the Rebana border.

What this means for the three workshop themes

1

Sustainable mobility & connectivity

Separate transit charging (toll, working) from resident charging (missing). Measure success as the share of residents within 10 minutes: from 29 % to at least 50 % needs about 12 sites.

2

Energy

Use substation headroom as a siting criterion. Haurgeulis, Cikedung, Arjawinangun Baru and Kadipaten can absorb 10–20 MW of chargers without reinforcement; towns need daytime time-of-use.

3

Skills

1.78 M jobs by 2030 and 10,000 ha of industrial land in west Subang will move demand to the north coast, where access is 19 %. Plan chargers, transformers and technician training by the KPI map.

Equity: spending-based concentration is close to zero, so a class gap has not formed yet. Kuningan and Sumedang, with no industrial zones, will be served last if siting follows the market.

Next steps and open data

Data that would sharpen the ex-ante analysis: monthly EV registrations by sub-district (Samsat), charger commissioning dates, and the 20 kV network plan for the industrial zones.

Report, slides and ur-scape package: jabar-ev.vercel.app/analysis/rebana_insights.html

Code and data: github.com/qastill/pathwaysevcharging